

## 1. What is Plotly?

**Plotly** is an open-source data visualization library used to create **interactive, high-quality graphs** in Python, R, and JavaScript.

In Python, it is widely used for:

- Interactive data visualization
- Dashboards
- Data exploration
- Web-based analytics applications

Unlike Matplotlib (which is mostly static), Plotly graphs are interactive by default:

- Zoom
- Pan
- Hover tooltips
- Toggle legend items
- Export as image

It works well in:

- Jupyter Notebook
- Google Colab
- Web applications
- Dash applications

## 2. Different Interfaces Available in Plotly (Python)

Plotly in Python mainly provides **two interfaces**:

### 1. Plotly Express (px)

- High-level interface
- Quick and simple plotting
- Less code
- Best for quick data exploration

### 2. Graph Objects (go)

- Low-level interface
- Full customization
- More control

- Used for complex and customized plots

There is also:

- **Dash** – for building interactive dashboards and web apps using Plotly.

### 3. Difference Between Plotly Express and Graph Objects

| Feature       | Plotly Express    | Graph Objects         |
|---------------|-------------------|-----------------------|
| Level         | High-level        | Low-level             |
| Code length   | Short             | Longer                |
| Customization | Limited           | Very detailed control |
| Ease of use   | Beginner friendly | Advanced              |
| Best for      | Quick analysis    | Production dashboards |

#### Example:

Plotly Express:

```
import plotly.express as px

fig = px.scatter(df, x="age", y="salary")

fig.show()
```

Graph Objects:

```
import plotly.graph_objects as go

fig = go.Figure()

fig.add_trace(go.Scatter(x=df["age"], y=df["salary"], mode='markers'))

fig.show()
```

### 4. How to Create Subplots in Plotly?

We use:

```
from plotly.subplots import make_subplots
```

Example:

```
from plotly.subplots import make_subplots

import plotly.graph_objects as go

fig = make_subplots(rows=1, cols=2)
```

```
fig.add_trace(go.Scatter(x=[1,2,3], y=[4,5,6]), row=1, col=1)
```

```
fig.add_trace(go.Bar(x=[1,2,3], y=[6,5,4]), row=1, col=2)
```

```
fig.show()
```

This allows:

- Multiple plots in one figure
- Shared axes
- Complex layout control

## 5. How to Export Plotly Charts?

### Export as Image (PNG, JPG, PDF)

```
fig.write_image("chart.png")
```

Requires:

```
pip install kaleido
```

### Export as HTML

```
fig.write_html("chart.html")
```

This preserves interactivity.

## 6. What is a Dashboard?

A **dashboard** is a visual interface that displays:

- Multiple charts
- KPIs
- Tables
- Filters
- Interactive components

All in one screen to monitor and analyze data.

Example uses:

- Sales monitoring
- Student performance tracking
- Hospital analytics
- Website traffic analysis

## 7. How to Make Interactive Dashboards?

In Python, we use:

### Dash (by Plotly)

Dash combines:

- Plotly (visualization)
- Flask (backend)
- React.js (frontend)

### Basic Structure of a Dash App:

```
from dash import Dash, dcc, html
```

```
import plotly.express as px
```

```
app = Dash(__name__)
```

```
fig = px.scatter(x=[1,2,3], y=[4,5,6])
```

```
app.layout = html.Div([  
    html.H1("My Dashboard"),  
    dcc.Graph(figure=fig)  
])
```

```
if __name__ == '__main__':  
    app.run_server(debug=True)
```

This creates a web-based interactive dashboard.

## 8. How to Create a Dashboard? (Steps)

1. Define objective (What problem?)
2. Identify KPIs
3. Prepare data
4. Create visualizations
5. Add filters/dropdowns
6. Arrange layout

## 7. Deploy

Deployment options:

- Local server
- Heroku
- AWS
- Render
- Azure

## 9. What is the Use of Heatmap?

A **heatmap** is used to visualize:

- Correlation matrix
- Density
- Intensity of values
- Pattern detection

Example uses:

- Correlation between features
- Student performance across subjects
- Website click density
- Hospital waiting times by hour/day

Example:

```
import plotly.express as px  
fig = px.imshow(corr_matrix)  
fig.show()
```

Heatmaps help detect:

- Strong relationships
- Clusters
- Outliers
- Trends

## **10. Different Types of Dashboards**

### **1. Operational Dashboard**

- Real-time monitoring
- Updated frequently
- Example: live website traffic

### **2. Analytical Dashboard**

- Used for deep analysis
- Includes filters, drill-downs
- Example: student performance analysis

### **3. Strategic Dashboard**

- High-level KPIs
- Used by top management
- Monthly/quarterly data

### **4. Tactical Dashboard**

- Department-level monitoring
- Focused insights
- Performance tracking

Plotly is:

- Interactive
- Web-friendly
- Easy to use
- Highly customizable

Plotly Express → Quick plotting

Graph Objects → Advanced control

Dash → Full interactive dashboards